

LAPORAN PRAKTIKUM
POSTTEST 2
JARINGAN KOMPUTER

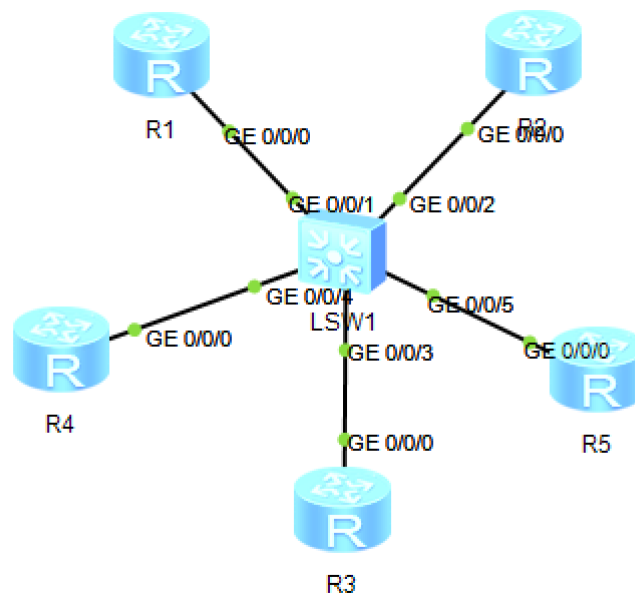


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1. Soal 1

Langkah pertama yang dilakukan adalah membuka workspace eNSP dan menambahkan 5 buah perangkat router dan 1 buah switch. Selanjutnya, hubungkan masing-masing router dengan switch menggunakan kabel jenis copper. Port GE0/0/0 pada router hubungkan dengan port GE0/0/1 pada switch dan begitupun untuk router lainnya (hanya port pada switch nya yang berbeda). Setelah kabel terpasang, blok seluruh topologi tersebut dan tekan tombol start hingga indikator port menyala.



2. Soal 2

Untuk mengganti nama router sesuai ketentuan pada tabel soal di classroom, konfigurasi dilakukan melalui CLI pada masing-masing router. Ketik `u t m` untuk menghilangkan log panjang yang akan muncul seiring melakukan konfigurasi. Ketik `system-view`, lalu `sysname server` (untuk R1). Setelah itu, pemberian ip address `10.10.10.(nim) 24` dengan masuk ke interface `g0/0/0` terlebih dahulu. Ip sudah tersimpan terlihat dari output setelah kita ketik `this`. Lalu `q` untuk keluar dari interfacenya. Lakukan hal yang sama ke semua router sesuai ketentuan hostnama dan ip addressnya.

```
R1
The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname server
[server]int g0/0/0
[server-GigabitEthernet0/0/0]ip address 10.10.10.67 24
[server-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
 ip address 10.10.10.67 255.255.255.0
#
return
[server-GigabitEthernet0/0/0]q
[server]|
```

```
R3
The device is running!

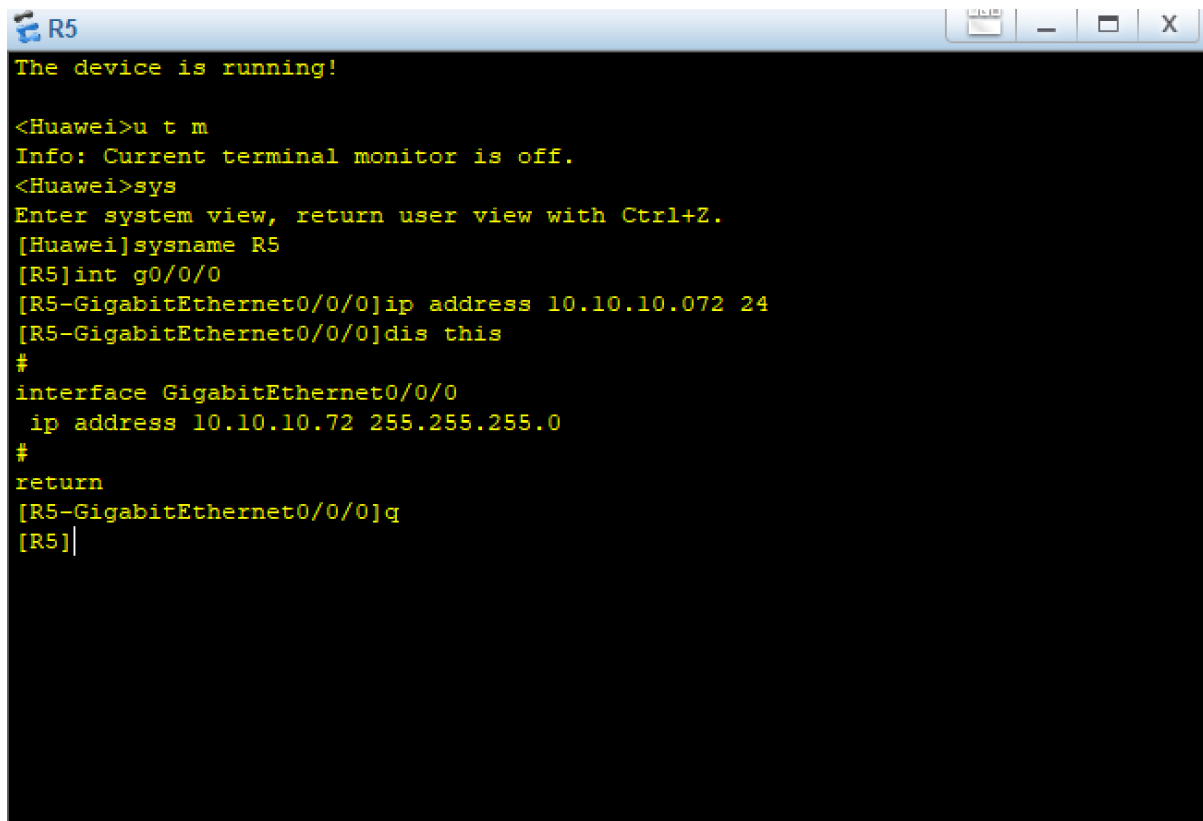
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname client
[client]int g0/0/0
[client-GigabitEthernet0/0/0]ip address 10.10.10.070 24
[client-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
 ip address 10.10.10.70 255.255.255.0
#
return
[client-GigabitEthernet0/0/0]q
[client]|
```

```
R2
The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.10.10.069 24
[R2-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
 ip address 10.10.10.69 255.255.255.0
#
return
[R2-GigabitEthernet0/0/0]q
[R2]|
```

```
R4
The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R4
[R4]int g0/0/0
[R4-GigabitEthernet0/0/0]ip address 10.10.10.071 24
[R4-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
 ip address 10.10.10.71 255.255.255.0
#
return
[R4-GigabitEthernet0/0/0]q
[R4]|
```

A screenshot of a terminal window titled 'R5'. The window has a blue title bar with standard window controls (minimize, maximize, close). The terminal text is as follows:

```
The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R5
[R5]int g0/0/0
[R5-GigabitEthernet0/0/0]ip address 10.10.10.072 24
[R5-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
 ip address 10.10.10.72 255.255.255.0
#
return
[R5-GigabitEthernet0/0/0]q
[R5]|
```

3. Soal 3

Pada soal ini, R1 digunakan menjadi ftp server. Konfigurasi dimulai dengan masuk ke mode otentikasi (aaa) untuk membuat kredensial pengguna lokal. Pengguna baru bernama mahdi ditambahkan dengan password Mahdi123, diberikan hak akses (privilege level) tertinggi yaitu level 3, dan tipe layanannya (service-type) dikhususkan untuk FTP. Direktori kerja untuk FTP diarahkan ke memori flash router (ftp-directory flash:). Terakhir, nyalakan layanan ftp dengan perintah ftp server enable.

```
R1
The device is running!

[server]aaa
[server-aaa]local-user mahdi password cipher Mahdi123
Info: Add a new user.
[server-aaa]local-user mahdi privilege level 3
[server-aaa]local-user mahdi ftp-directory flash:
[server-aaa]local-user mahdi service-type ftp
[server-aaa]dis local-user
-----
User-name                State  AuthMask  AdminLevel
-----
admin                    A      H          -
mahdi                    A      F          3
-----
Total 2 user(s)
[server-aaa]q
[server]ftp server enable
Info: Succeeded in starting the FTP server.
[server]q
<server>
```

4. Soal 4

Setelah melakukan konfigurasi ftp, simpan konfigurasi R1 (server) ke dalam file bernama server(nim).zip, sementara pada R3 yang bertindak sebagai klien, konfigurasinya disimpan ke dalam file bernama client(nim).zip. Setelah proses penyimpanan dikonfirmasi dengan mengetikkan 'y', perintah dir digunakan untuk melihat isi direktori flash dan memastikan bahwa file arsip berekstensi .zip tersebut telah benar-benar terbentuk dan tersimpan di dalam sistem.

```
R1
The device is running!

<server>save server067.zip
Are you sure to save the configuration to flash:/server067.zip?[Y/N]:y
Now saving the current configuration to the slot 17.
Save the configuration successfully.
<server>dir
Directory of flash:/

   Idx  Attr      Size(Byte)  Date       Time        FileName
   ---  ---      -
   0   drw-          -   Aug 07 2015 13:51:14   src
   1   drw-          -   Sep 18 2026 19:08:01   pmdata
   2   drw-          -   Sep 18 2026 19:08:07   dhcp
   3   -rw-         28   Sep 18 2026 19:08:08   private-data.txt
   4   drw-          -   Sep 18 2026 19:23:12   mplstpoam
   5   -rw-        532   Sep 18 2026 19:27:59   server067.zip

32,004 KB total (31,993 KB free)

<server>
```

```
R3
The device is running!

<client>save client067.zip
Are you sure to save the configuration to flash:/client067.zip?[Y/N]:y
Now saving the current configuration to the slot 17.
Sep 18 2026 19:29:28-08:00 client %%01CFM/4/SAVE_FILE(1)[0]:When deciding whether
to save the configuration to the file flash:/client067.zip, the user chose Y.
Save the configuration successfully.
<client>dir
Directory of flash:/

   Idx  Attr      Size(Byte)  Date       Time        FileName
   ---  ---      -
   0   drw-          -   Aug 07 2015 13:51:14   src
   1   drw-          -   Sep 18 2026 19:08:03   pmdata
   2   drw-          -   Sep 18 2026 19:08:10   dhcp
   3   -rw-         28   Sep 18 2026 19:08:10   private-data.txt
   4   drw-          -   Sep 18 2026 19:23:15   mplstpoam
   5   -rw-        452   Sep 18 2026 19:29:28   client067.zip

32,004 KB total (31,993 KB free)

<client>|
```

5. Soal 5

Langkah ini menunjukkan bagaimana R3 (klien) mentransfer file konfigurasinya ke R1 (server) melalui layanan FTP. Dari R3, koneksi FTP dilakukan ke alamat server 10.10.10.067 menggunakan username mahdi dan password yang telah dibuat sebelumnya. Setelah berhasil login, mode transfer data diubah menjadi binary agar file arsip tidak rusak saat dikirim. File konfigurasi klien kemudian diunggah ke server menggunakan perintah put client067.zip. Pengecekan menggunakan perintah dir di dalam sesi FTP membuktikan bahwa file client067.zip dan server067.zip sekarang berada pada directory di R1.

```
R3
The device is running!

<client>ftp 10.10.10.067
Trying 10.10.10.067 ...
Press CTRL+K to abort
Connected to 10.10.10.067.
220 FTP service ready.
User(10.10.10.067:(none)):mahdi
331 Password required for mahdi.
Enter password:
230 User logged in.

[ftp]binary
200 Type set to I.

[ftp]put client067.zip
200 Port command okay.
150 Opening BINARY mode data connection for client067.zip.

100%
226 Transfer complete.
FTP: 452 byte(s) sent in 0.310 second(s) 1.45Kbyte(s)/sec.

[ftp]dir
200 Type set to A.
```

```
[ftp]dir
200 Type set to A.

200 Port command okay.
150 Opening ASCII mode data connection for *.
drwxrwxrwx  1 noone  nogroup    0 Aug 07  2015 src
drwxrwxrwx  1 noone  nogroup    0 Sep 18 19:08 pmdata
drwxrwxrwx  1 noone  nogroup    0 Sep 18 19:08 dhcp
-rwxrwxrwx  1 noone  nogroup   28 Sep 18 19:08 private-data.txt
drwxrwxrwx  1 noone  nogroup    0 Sep 18 19:23 mplstpam
-rwxrwxrwx  1 noone  nogroup   452 Sep 18 19:30 client067.zip
-rwxrwxrwx  1 noone  nogroup   532 Sep 18 19:27 server067.zip
226 Transfer complete.
200 Type set to I.

[ftp]
```

6. Soal 6

Langkah ini untuk mereplikasi konfigurasi R3 (klien) dari R1 ke R4 dan R5. Masing-masing router (R4 dan R5) melakukan koneksi FTP ke R1 (10.10.10.067) untuk mengunduh file client067.zip menggunakan perintah get client067.zip. Setelah file tersebut terunduh ke memori flash perangkat, file zip tersebut ditetapkan sebagai file konfigurasi utama yang akan dibaca saat booting menggunakan perintah startup saved-configuration client067.zip. Setelah router di-reboot, nama sistem perangkat tersebut otomatis berubah menjadi client yang menandakan bahwa seluruh konfigurasi (termasuk hostname) dari file zip R3 telah sukses diimplementasikan secara identik.

```
R4
The device is running!

<R4>ftp 10.10.10.067
Trying 10.10.10.067 ...
Press CTRL+K to abort
Connected to 10.10.10.067.
220 FTP service ready.
User(10.10.10.067:(none)):mahdi
331 Password required for mahdi.
Enter password:
230 User logged in.

[ftp]binary
200 Type set to I.

[ftp]get client067.zip
Warning: The file client067.zip already exists. Overwrite it? [Y/N]:y
200 Port command okay.
150 Opening BINARY mode data connection for client067.zip.

226 Transfer complete.
FTP: 452 byte(s) received in 0.390 second(s) 1.15Kbyte(s)/sec.

[ftp]q
221 Server closing.

<R4>startup saved-configuration client067.zip
Info: Succeeded in setting the configuration for booting system.
<R4>reboot
Info: The system is now comparing the configuration, please wait.
Warning: All the configuration will be saved to the configuration file for the
ext startup:flash:/client067.zip, Continue?[Y/N]:n
Info: If want to reboot with saving diagnostic information, input 'N' and then
xecute 'reboot save diagnostic-information'.
System will reboot! Continue?[Y/N]:y
Sep 18 2026 19:47:32-08:00 R4 %01CMD/4/REBOOT(1)[0]:The user chose Y when dec
ing whether to reboot the system. (Task=co0, Ip=**, User=**)
<R4>
```

```
R4
The device is running!

<client>dir
Directory of flash:/

  Idx  Attr      Size(Byte)  Date       Time       FileName
  ---  ---
    0  drw-          -   Aug 07 2015 13:51:14   src
    1  drw-          -   Sep 18 2026 19:08:03   pmdata
    2  drw-          -   Sep 18 2026 19:08:09   dhcp
    3  -rw-         28   Sep 18 2026 19:47:49   private-data.txt
    4  drw-          -   Sep 18 2026 19:23:14   mplstpam
    5  -rw-        452   Sep 18 2026 19:40:22   client067.zip

32,004 KB total (31,993 KB free)

<client>|
```

```
R5
The device is running!

<R5>ftp 10.10.10.067
Trying 10.10.10.067 ...
Press CTRL+K to abort
Connected to 10.10.10.067.
220 FTP service ready.
User(10.10.10.067:(none)):mahdi
331 Password required for mahdi.
Enter password:
230 User logged in.

[ftp]binary
200 Type set to I.

[ftp]get client067.zip
200 Port command okay.
150 Opening BINARY mode data connection for client067.zip.

226 Transfer complete.
FTP: 452 byte(s) received in 0.320 second(s) 1.41Kbyte(s)/sec.

[ftp]q

221 Server closing.

<R5>startup saved-configuration client067.zip
Info: Succeeded in setting the configuration for booting system.
<R5>reboot
Info: The system is now comparing the configuration, please wait.
Warning: All the configuration will be saved to the configuration file for the
ext startup:flash:/client067.zip, Continue?[Y/N]:n
Info: If want to reboot with saving diagnostic information, input 'N' and then
execute 'reboot save diagnostic-information'.
System will reboot! Continue?[Y/N]:y
Sep 18 2026 19:51:34-08:00 R5 %%01CMD/4/REBOOT(1)[0]:The user chose Y when dec
iding whether to reboot the system. (Task=co0, Ip=**, User=**)
<R5>
```

```
R5
The device is running!

<client>dir
Directory of flash:/

   Idx  Attr      Size(Byte)  Date       Time       FileName
   ---  ---      -          -          -          -
   0    drw-          -    Aug 07 2015 13:51:14    src
   1    drw-          -    Sep 18 2026 19:08:03    pmdata
   2    drw-          -    Sep 18 2026 19:08:09    dhcp
   3    -rw-         28    Sep 18 2026 19:51:49    private-data.txt
   4    drw-          -    Sep 18 2026 19:23:15    mplstpoam
   5    -rw-        452    Sep 18 2026 19:50:23    client067.zip

32,004 KB total (31,993 KB free)

<client>
```

7. Soal 7

Sama seperti langkah sebelumnya, tetapi kali ini mereplikasi konfigurasi R1 (server) dari R1 ke R2. R2 login melalui FTP ke R1 (10.10.10.067) lalu mengambil file server067.zip dengan perintah `get server067.zip`. File tersebut kemudian dideklarasikan sebagai file startup dengan mengeksekusi startup saved-configuration server067.zip. Router R2 kemudian di-reboot. Hasilnya, setelah proses reboot selesai, hostname R2 berubah menjadi server yang menandakan bahwa seluruh konfigurasi (termasuk hostname) dari file zip R1 telah sukses diimplementasikan secara identik.

```
R2
The device is running!

<R2>ftp 10.10.10.067
Trying 10.10.10.067 ...
Press CTRL+K to abort
Connected to 10.10.10.067.
220 FTP service ready.
User(10.10.10.067:(none)):mahdi
331 Password required for mahdi.
Enter password:
230 User logged in.

[ftp]binary
200 Type set to I.

[ftp]get server067.zip
200 Port command okay.
150 Opening BINARY mode data connection for server067.zip.

226 Transfer complete.
FTP: 532 byte(s) received in 0.350 second(s) 1.52Kbyte(s)/sec.

[ftp]q

221 Server closing.

<R2>startup saved-configuration server067.zip
Info: Succeeded in setting the configuration for booting system.
<R2>reboot
Info: The system is now comparing the configuration, please wait.
Warning: All the configuration will be saved to the configuration file for the
ext startup:flash:/server067.zip, Continue?[Y/N]:n
Info: If want to reboot with saving diagnostic information, input 'N' and then
execute 'reboot save diagnostic-information'.
System will reboot! Continue?[Y/N]:y
Sep 18 2026 19:55:31-08:00 R2 %%01CMD/4/REBOOT(1)[0]:The user chose Y when dec
ing whether to reboot the system. (Task=co0, Ip=**, User=**)
<R2>
```

```
R2
The device is running!

<server>dir
Directory of flash:/

  Idx  Attr      Size(Byte)  Date       Time       FileName
  ---  ---
    0  drw-          -   Aug 07 2015 13:51:14   src
    1  drw-          -   Sep 18 2026 19:08:03  pmdata
    2  drw-          -   Sep 18 2026 19:08:09  dhcp
    3  -rw-         28   Sep 18 2026 19:55:46  private-data.txt
    4  drw-          -   Sep 18 2026 19:23:15  mplstpoam
    5  -rw-        532   Sep 18 2026 19:54:30  server067.zip

32,004 KB total (31,993 KB free)

<server>
```